

Equivalent fractions



1 Select the equivalent fractions from the following lists:

a $\frac{4}{40}$, $\frac{1}{10}$, $\frac{2}{80}$, $\frac{8}{80}$, $\frac{8}{20}$

c $\frac{1}{4}$, $\frac{16}{64}$, $\frac{4}{8}$, $\frac{128}{256}$, $\frac{32}{64}$

e $\frac{15}{20}$, $\frac{6}{8}$, $\frac{10}{15}$, $\frac{9}{12}$, $\frac{2}{3}$

g $\frac{4}{18}$, $\frac{14}{63}$, $\frac{2}{16}$, $\frac{2}{9}$, $\frac{10}{36}$

b $\frac{20}{30}$, $\frac{7}{3}$, $\frac{8}{12}$, $\frac{2}{6}$, $\frac{2}{3}$

d $\frac{9}{8}$, $\frac{27}{24}$, $\frac{108}{96}$, $\frac{8}{9}$, $\frac{54}{24}$

f $\frac{6}{9}$, $\frac{3}{4}$, $\frac{10}{15}$, $\frac{17}{24}$, $\frac{4}{6}$

h $\frac{17}{5}$, $\frac{8}{20}$, $\frac{2}{5}$, $\frac{4}{14}$, $\frac{16}{40}$

2 Complete the following equivalent fraction statements:

a $\frac{1}{2} = \frac{\square}{4}$

b $\frac{1}{5} = \frac{\square}{75}$

c $\frac{3}{8} = \frac{\square}{40}$

d $\frac{9}{5} = \frac{\square}{40}$

e $\frac{19}{10} = \frac{\square}{90}$

f $\frac{3}{8} = \frac{\square}{24}$

g $\frac{5}{16} = \frac{20}{\square}$

h $\frac{7}{12} = \frac{35}{\square}$

i $\frac{4}{9} = \frac{40}{\square}$

j $\frac{4}{3} = \frac{48}{\square}$

k $\frac{5}{21} = \frac{45}{\square}$

l $\frac{18}{11} = \frac{72}{\square}$

3 Complete the table below:

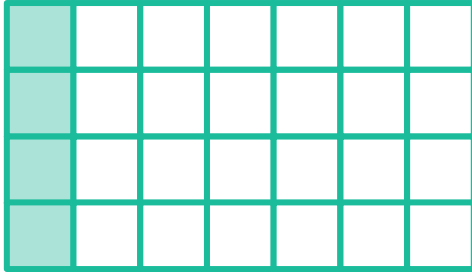
Mixed number	Improper fraction	Equivalent fraction
$2\frac{1}{6}$		
$3\frac{4}{5}$		
$4\frac{2}{7}$		
$1\frac{2}{5}$		
$6\frac{1}{8}$		
$2\frac{3}{8}$		
$3\frac{3}{7}$		
$4\frac{2}{9}$		
$5\frac{1}{4}$		



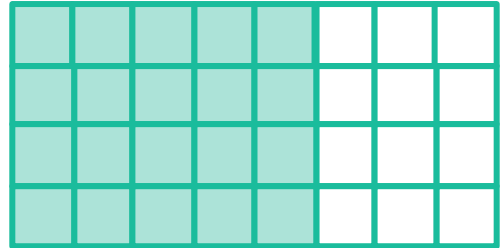
Simplify fractions

 4 Use the diagrams provided to simplify the following fractions:

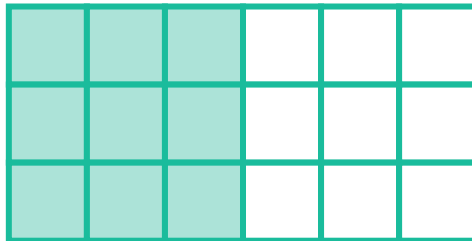
a $\frac{4}{28}$



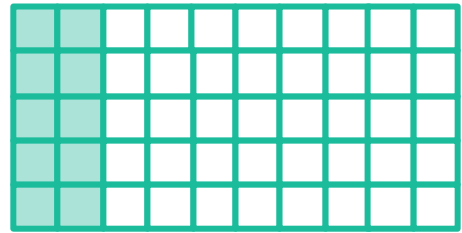
b $\frac{20}{32}$



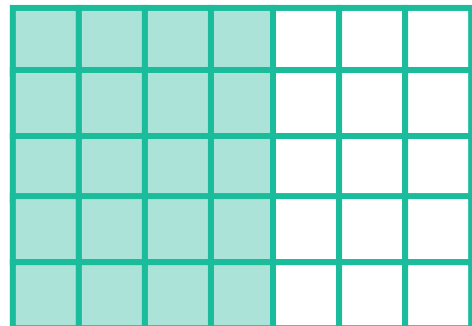
c $\frac{9}{18}$



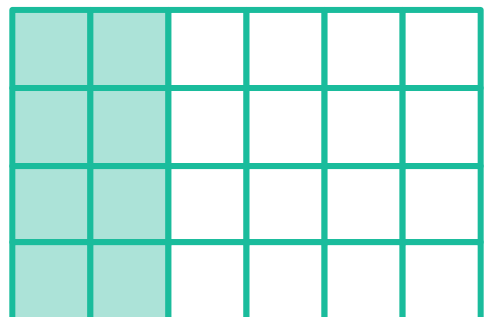
d $\frac{10}{50}$



e $\frac{20}{35}$



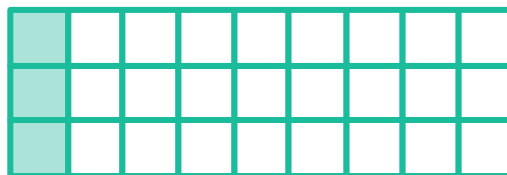
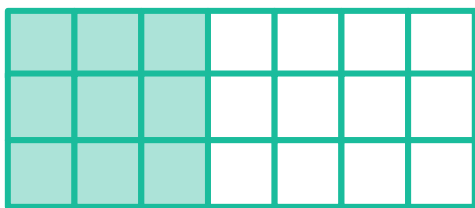
f $\frac{8}{24}$



$$g \quad \frac{9}{21}$$



$$h \quad \frac{3}{27}$$



5 Simplify the following fractions:

$$a \quad \frac{3}{9}$$

$$b \quad \frac{20}{60}$$

$$c \quad \frac{6}{20}$$

$$d \quad \frac{27}{45}$$

$$e \quad \frac{63}{28}$$

$$f \quad \frac{40}{30}$$

$$g \quad \frac{26}{50}$$

$$h \quad \frac{24}{66}$$

$$i \quad \frac{32}{80}$$

$$j \quad \frac{64}{92}$$

$$k \quad \frac{80}{100}$$

$$l \quad \frac{60}{200}$$

$$m \quad \frac{2}{10}$$

$$n \quad \frac{8}{64}$$

$$o \quad \frac{21}{231}$$

$$p \quad \frac{150}{270}$$

Compare fractions

6 For the each of the following pairs of fractions, state which fraction is bigger:

$$a \quad \frac{1}{3}, \quad \frac{3}{15}$$

$$b \quad \frac{4}{28}, \quad \frac{6}{14}$$

$$c \quad \frac{31}{8}, \quad \frac{24}{7}$$

$$d \quad \frac{3}{8}, \quad \frac{1}{2}$$

$$e \quad \frac{14}{5}, \quad \frac{11}{4}$$

$$f \quad \frac{5}{2}, \quad \frac{49}{20}$$

$$g \quad 1\frac{6}{7}, \quad \frac{25}{14}$$

$$h \quad 3\frac{4}{9}, \quad \frac{7}{2}$$

7 For the each of the following pairs of fractions, state which fraction is smaller:

$$a \quad \frac{2}{9}, \quad \frac{7}{8}$$

$$b \quad \frac{7}{5}, \quad \frac{8}{10}$$

$$c \quad \frac{4}{15}, \quad \frac{2}{5}$$

$$d \quad \frac{4}{7}, \quad \frac{4}{8}$$

$$e \quad \frac{63}{15}, \quad \frac{82}{20}$$

$$f \quad \frac{21}{6}, \quad \frac{18}{5}$$

$$g \quad \frac{27}{45}, \quad \frac{5}{8}$$

$$h \quad 2\frac{5}{9}, \quad 2\frac{7}{11}$$

8 For each of the following groups of fractions



- i Find the lowest common denominator of the 3 fractions.
- ii Rewrite all 3 fractions with this lowest common denominator.
- iii List the 3 fractions in ascending order.

a $\frac{2}{5}, \frac{9}{55}, \frac{7}{11}$

b $\frac{7}{13}, \frac{8}{65}, \frac{3}{5}$

c $\frac{5}{7}, \frac{9}{77}, \frac{8}{11}$

d $\frac{5}{17}, \frac{7}{68}, \frac{3}{4}$

9 Arrange the following groups of fractions from smallest to largest:

a $\frac{11}{20}, \frac{4}{15}, \frac{2}{5}$

b $\frac{2}{3}, \frac{3}{7}, \frac{4}{5}$

c $\frac{21}{9}, \frac{6}{4}, \frac{2}{7}$

d $4\frac{1}{2}, \frac{31}{4}, \frac{15}{2}$

e $\frac{39}{4}, 4\frac{5}{10}, \frac{19}{2}$

f $1\frac{6}{7}, \frac{33}{14}, \frac{16}{7}$

g $2\frac{2}{5}, \frac{33}{10}, \frac{16}{5}$

h $\frac{17}{3}, \frac{35}{6}, 3\frac{2}{3}$

10 For each of the following groups of fractions:

- i Find the lowest common denominator of the 3 fractions.
- ii Rewrite all 3 fractions with this lowest common denominator.
- iii List the 3 fractions in descending order.

a $\frac{1}{4}, \frac{3}{8}, \frac{5}{32}$

b $\frac{2}{6}, \frac{4}{9}, \frac{1}{54}$

c $\frac{2}{5}, \frac{6}{7}, \frac{8}{35}$

d $2\frac{2}{5}, \frac{37}{10}, \frac{18}{5}$

11 Arrange the following groups of fractions from largest to smallest:

a $\frac{1}{2}, \frac{2}{3}, \frac{5}{8}$

b $\frac{3}{8}, \frac{9}{16}, \frac{13}{24}$

c $2\frac{7}{10}, \frac{13}{5}, \frac{7}{2}$

d $2\frac{1}{7}, \frac{33}{14}, \frac{16}{7}$

e $6\frac{1}{2}, \frac{35}{4}, \frac{17}{2}$

f $\frac{33}{10}, 1\frac{4}{5}, \frac{16}{5}$

g $\frac{17}{4}, 4\frac{3}{4}, \frac{39}{8}$

h $\frac{21}{2}, \frac{37}{4}, 9\frac{1}{2}$